



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognized by AICTE, New Delhi)
Accredited by NAAC with 'A' Grade, All UG Programmes are Accredited by NBA
CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

ELECTRICAL & ELECTRONICS ENGINEERING

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH

I-SEMESTER

(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 EE 3101	Electrical Machines-II	PC	3	3	--	--	25	75	100
B19 EE 3102	Control Systems	PC	3	3	--	--	25	75	100
B19 EE 3103	Power Electronics	PC	3	3	--	--	25	75	100
B19 EE 3104	Electrical Power Generation, Transmission & Distribution	PC	3	3	--	--	25	75	100
B19 EE 3105	Micro Processors and Micro Controllers	PC	3	3	--	--	25	75	100
# PE-I	Professional Elective-I	PE	3	3	--	--	25	75	100
B19 EE 3110	Electrical Machines-I Lab	PC	1.5	--	--	3	20	30	50
B19 EE 3111	Microprocessor & Microcontroller Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3101	Employability skills-1	MC	--	3	--	--	--	--	--
B19 MC 3102	Basic Coding	MC	--	--	--	3	--	--	--
TOTAL			21	21	--	9	190	510	700

#PE-I	Course Code	Course
	B19 EE 3106	Analog Electronics
	B19 EE 3107	Linear and Digital IC Applications
	B19 EE 3108	Pulse and Digital Circuits
	B19 EE 3109	MOOCs-I

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3101	PC	3	--	--	3	25	75	3 Hrs.
ELECTRICAL MACHINES-II								
(For EEE)								
Course Objectives: Students will learn								
1.	The concept of synchronous generator, EMF equation and air-gap MMF distribution.							
2.	Construction of synchronous machine, voltage regulation, power angle characteristics of salient and non-salient pole synchronous generator.							
3.	The operation and starting methods of synchronous motor, operating characteristics, V-curves							
4.	construction of three phase induction motor, speed-torque characteristics, circle diagram							
5.	Starting and speed control of 3- ϕ induction motor, double revolving field theory, equivalent circuit, starting method of 1- ϕ induction motors.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge level
1.	Illustrate the constructional features of AC rotating machines, MMF distribution in air-gap, Rotating Magnetic field and calculate generated voltage.							K3
2.	Compute voltage regulation and analyze the power angle characteristics and explain parallel operation of salient pole synchronous generator.							K3, K4
3.	Analyze characteristics of salient pole synchronous motor and explain the starting methods of synchronous motor.							K3, K4
4.	Illustrate the operation of three phase induction motor and apply the slip-torque equations in Analyzing the performance of 3 Φ Induction Motor.							K3, K4
5.	Demonstrate the starting and speed control of 3- Φ induction motor, operation of single-phase induction motor and its starting methods.							K3
SYLLABUS								
UNIT-I (8Hrs)	BASIC CONCEPTS IN AC ROTATING MACHINES: Constructional features- cylindrical rotor, Salient pole rotor, single turn coil - active portion and overhang; full-pitch coils, concentrated winding, generated EMF, distributed winding, Harmonic content in distributed winding, Short pitched coils (Simple Problems), Air-gap MMF of Distributed AC windings-MMF space wave of single coil, MMF space wave of one phase of distributed winding, Production of Rotating Magnetic Field (only concept).							
UNIT-II (12 Hrs)	SYNCHRONOUS GENERATORS: Circuit model, armature reaction, Determination of synchronous Reactance-OCC and SCC, voltage regulation, voltage regulation by EMF, MMF and ZPF methods, Salient pole Machine -two reaction theory, analysis of phasor diagram, power angle characteristics, determination of x_d and x_q (Simple Problems), Synchronizing to infinite bus, Parallel operation of Alternators (Only concepts).							
UNIT-III (10 Hrs)	SYNCHRONOUS MOTORS: Operating principle, starting methods of synchronous motors, salient pole synchronous motor-Analysis of phasor diagram (simple problems), Operating characteristics of synchronous motor-variable load and constant excitation, V-curves, variable excitation and constant load.							

UNIT-IV (12Hrs)	INDUCTION MACHINES Construction, Types (squirrel cage and slip-ring), Equivalent circuit- Starting and Maximum Torque, Torque Slip Characteristics, Effect of parameter variation on torque speed characteristics (variation of rotor resistance), No-load and Blocked rotor test, Losses and Efficiency, Circle Diagram.
UNIT-V (8 Hrs)	STARTING AND SPEED CONTROL OF 3-Φ INDUCTION MOTOR AND 1-Φ INDUCTION MOTORS Methods of starting, braking and speed control for 3- Φ induction motors (simple problems). Self-excited Induction Generator. 1- ϕ induction motors-constructional features, double revolving field theory, equivalent circuit, determination of parameters (only concept), Split phase starting methods & applications.
Text Books:	
1.	Nagrath & Kothari, "Electric Machines" TMH 5th edition
2.	PS Bhimbra, "Electrical Machinery", Khanna Publishers. 7th edition.
Reference Books:	
1.	Fitzgerald & Kingsley, "Electric Machinery" McGraw Hill, 6 th edition

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3102	PC	3	--	--	3	25	75	3 Hrs.
CONTROL SYSTEMS								
(For EEE)								
Course Objectives: Students will learn								
1.	The modeling of linear systems using transfer functions and obtain transfer functions for physical electrical and mechanical systems.							
2.	To represent systems using block diagrams and signal flow graphs and derive their transfer functions.							
3.	The significance of time response and find it for system analysis in transient and steady state.							
4.	The concept of stability and root locus and know different techniques of stability analysis.							
5.	The concept of frequency response and its application for control system analysis.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Derive transfer functions for electrical and mechanical physical systems by applying laws of physics.							K3
2.	Prepare block diagrams & Signal Flow Graphs for systems and find their transfer functions.							K3
3.	Analyze systems in time domain for transient and steady-state behavior							K3, K4
4.	Ascertain the stability of systems using RH criterion and Root locus.							K3
5.	Illustrate the frequency response plots of systems and use them for system analysis and stability assessment.							K3, K4
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to control systems: Open loop and closed loop systems- Transfer Functions of Linear Systems– Impulse Response of Linear Systems – Mathematical Modeling of Physical Systems – Equations of Electrical Networks – Modeling of Mechanical Systems – Equations of Mechanical Systems-Analogous Systems.							
UNIT-II (10 Hrs)	Block Diagrams and Signal Flow Graphs: Block Diagrams of Control Systems – Signal Flow Graphs (Simple Problems) – Reduction Techniques for Complex Block Diagrams and Signal Flow Graphs (Simple Examples)-Feedback Characteristics of Control Systems							
UNIT-III (10 Hrs)	Time Domain Analysis: Time Domain Analysis of Control System-Time Response of First and Second Order Systems with Standard Input Signals – Steady State Error Constants – Effect of Derivative and Integral Control on Transient and Steady State Performance of Feedback Control Systems.							
UNIT-IV (10 Hrs)	Stability of Control Systems:Concept of Stability– Routh-Hurwitz Criterion, Relative Stability Analysis, the Concept and Construction of Root Loci, Analysis of Control Systems with Root Locus (Simple Problems to understand theory).							

UNIT-V (10 Hrs)	Frequency Domain Analysis: Frequency Domain Analysis of control systems - Bode Plots- Log Magnitude versus Phase Plots- Polar Plots -Correlation between Time and Frequency Responses - Nyquist Stability Criterion -Assessment of Relative Stability -All Pass and Minimum Phase Systems -Constant M and N Circles.
Text Books:	
1.	I. J. Nagrath and M. Gopal, ‘Control Systems Engineering’, New Age International Publishers (6th Edition).
2.	Norman S. Nise, ‘Control Systems Engineering’, Wiley publications (8 th Edition)
Reference Books:	
1.	Benjamin C. Kuo, ‘Automatic Control Systems’, PHI (5th Edition).
2.	Richard C. Dorf and Robert H. Bishop, ‘Modern Control Systems’, Addison-Wesley Publishers (8th Edition)

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3103	PC	3	--	--	3	25	75	3 Hrs.
POWER ELECTRONICS								
(For EEE)								
Course Objectives: Students will learn								
1.	Different power semiconductor devices, series and parallel connection of SCR's, protection circuits commutation and triggering of SCR's.							
2.	The concepts of Phase controlled rectifiers with different loads.							
3.	Different modes of operation of DC-DC Choppers.							
4.	Cycloconverter and single-phase AC voltage controllers.							
5.	Inverters different mode of operation and harmonic reduction techniques.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate different Power Semi-conductor devices and Analyze Series and Parallel connections of SCR's.							K4, K3
2.	Analyze the phase-controlled rectifiers with different loads.							K4
3.	Explain DC-DC converters for power electronic applications.							K3
4.	Acquire knowledge on single phase AC voltage controllers.							K3
5.	Analyze the operation of inverters.							K4
SYLLABUS								
UNIT-I (10 Hrs)	POWER SEMI CONDUCTOR DEVICES: Power Diode - Silicon Controlled Rectifiers (SCRs) – Power MOSFET – Power IGBT and their static characteristics. Dynamic characteristics of SCR – Turn on and turn off methods. Two transistor analogy of SCR -Series and parallel connections of SCRs- Numerical problems - Snubber circuit details –. SCR trigger circuits-R, RC and UJT triggering circuits - Impulse Commutation-Resonant pulse commutation-Complementary commutation.							
UNIT-II (10 Hrs)	PHASE CONTROLLED RECTIFIERS Principles of phase-controlled rectification -Study of Single phase and three-phase half controlled and full controlled bridge rectifiers with R, RL, RLE loads - Numerical problems. Effect of source inductance.							
UNIT-III (10 Hrs)	CHOPPERS Principle of operation and Design of Buck, Boost, Buck-Boost and Cuk converters continuous mode of operation - numerical problems.							
UNIT-IV (10 Hrs)	1-Ø AC-AC CONVERTERS: Principle of operation of Single-phase Cycloconverter continuous and discontinuous Single phase AC Voltage Controllers with R and RL loads.							

UNIT-V (10 Hrs)	INVERTERS: Principle of operation of Single-phase Inverters - Three phase Inverters (180^0 and 120^0 modes) - Comparison of Voltage Source Inverters and Current source Inverters-Harmonic reduction techniques, Harmonic reduction by PWM, Harmonic reduction by transformer connections, Harmonic reduction by steeped wave –PWM ,Single pulse width, Multipulse width, sinusoidal pulse width.
Text Books:	
1.	Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3rd edition.
2.	Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition
3.	Power Electronics by Daniel W Hart-Mc Graw Hill Publications,1st Edition.
Reference Books:	
1.	Power Electronics – VedamSubramanyam, New Age International (p) Limited, Publishers.
2.	Power Electronics – P.C. Sen, Tata Mc Graw-Hill Publishing.
3.	Thyristorised power Controllers – G.K. Dubey, S.R Doradra, A. Joshi and R.M.K. Sinha, New Age international Pvt Ltd. Publishers latest edition.
4.	Power electronics – M.D. Singh & K.B. Kanchandhani, Tata Mc Graw – Hill Publishing Company, 2nd edition.
Web links:	
1.	https://www.youtube.com/watch?v=1Auay7ja2oY&list=PLA07ACBDE053A8229
2.	https://www.youtube.com/watch?v=ZbvWe9xBu3Q&list=PLp6ek2hDcoND7i5-DAD9mPmYF1Wg6ROdO

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3104	PC	3	--	--	3	25	75	3 Hrs.
ELECTRICAL POWER GENERATION, TRANSMISSION & DISTRIBUTION								
(For EEE)								
Course Objectives: Students will learn								
1.	Understand the general arrangement, principles and components & their functions present in thermal, hydro, nuclear and gas power plants.							
2.	Understand the load curves and different types of tariffs.							
3.	Know the performance analysis of transmission lines							
4.	Give emphasis on mechanical design of transmission line conductors and insulators.							
5.	Study the different types of distribution systems							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Describe the power generation from different energy sources, tariffs and Economic aspects.							K3
2.	Apply Kelvin’s law and analyze different type’s transmission and distribution networks.							K3, K4
3.	Calculate Inductance & Capacitance of transmission lines							K3
4.	Determine the performance of short, medium and long transmission lines.							K3
5.	Explain the mechanical and electrical design aspects of transmission system							K3
SYLLABUS								
UNIT-I (10 Hrs)		ELECTRIC POWER GENERATION & ECONOMIC CONSIDERATIONS: Layout of thermal, hydro, nuclear and gas power plants, brief description of various parts of different power plants. Load curves and associated definitions, load duration curves, different types of tariffs and examples.						
UNIT-II (12 Hrs)		POWER SUPPLY SYSTEMS& DISTRIBUTION SYSTEMS: Transmission and distribution systems- D.C 2-wire and 3- wire systems, A.C single phase, three phase and 4-wire systems, comparison of copper efficiency. Primary and secondary distribution systems concentrated & uniformly distributed loads on distributors fed at both ends, ring distributor, voltage drop and power loss calculation, Kelvin’s law.						
UNIT-III (10 Hrs)		INDUCTANCE & CAPACITANCE CALCULATIONS: Types of conductors, line parameters, calculation of inductance and capacitance of single and double circuit transmission lines, three phase lines with bundle conductors. Skin effect and proximity effect.						
UNIT-IV (9 Hrs)		PERFORMANCE OF TRANSMISSION LINES: Generalized network constants and equivalent circuits of short, medium, long transmission line. Line performance: regulation and efficiency, Ferranti effect.						

UNIT-V (9 Hrs)	OVERHEAD LINE INSULATORS: Types of insulators, potential distribution over a string of suspended insulators, methods of equalizing potential. Corona: phenomenon of corona, corona loss, concept of radio interference. MECHANICAL DESIGN OF TRANSMISSION LINES: Different types of tower, sag –tension calculations, sag template, string charts.
Text Books:	
1.	Wadhwa,C.L., “ Electric Power Systems”, New Age International Private Limited; Sixth edition (2010).
2.	Power System Analysis and Design by Dr. B.R Gupta S Chand &Company; 2005.
3.	Nagarath,I.J, and Kothari, D.P., “Power System Engineering”, McGraw Hill Education; 2 edition (2007).
4.	“A Course in PowerSytems” by J.B Gupta, S.K. Kataria& Sons; 2013 edition.
Reference Books:	
1.	Burke James,J., “Power Distribution Engineering; Fundamentals and Applications “ Marcel Dekker 1996.
2.	Grainger john, J. and Stevenson,Jr. W.D., “Power System Analysis”, McGraw Hill,1994.
3.	Harder Edwin,I., “Fundamentals of Energy Production”,JohnWiely and Sons,1982.
4.	Deshpande, M.V., “Elements of Electric Power Station Design”,A.H Wheeler andCo.Allahabad,1979.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3105	PC	3	--	--	3	25	75	3 Hrs.
MICRO PROCESSORS AND MICRO CONTROLLERS								
(For EEE)								
Course Objectives: Students will learn								
1.	Architecture of 8085 microprocessor, Addressing modes, and simple Programmes.							
2.	Architecture of 8086 microprocessor, Addressing modes, and simple Programmes.							
3.	Architecture of different peripherals like 8259,8255,8279 and A/D and D/A converters and interfacing							
4.	Architecture of 8051 microcontroller, Memory Organization, Timers/Counters, Serial communication, Interrupts.							
5.	Addressing Modes, Instruction set of 8051 and simple interfacing applications development with programming 8051							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate the Architecture, Addressing modes, and Memory organization of 8085 Microprocessor.							K3
2.	Illustrate the Architecture, Addressing modes, and Memory organization of 8086 Microprocessor.							K3
3.	Illustrate how the different peripherals (8255, 8253 etc.) are interfaced with Microprocessor.							K3
4.	Illustrate the Architecture, Memory organization, Timers/Counters, Serial Communication and Interrupts of 8051 Microprocessor.							K3
5.	List out the 8051 instructions and develop programming to interface 8051 microcontrollers for simple real time applications.							K4
SYLLABUS								
UNIT-I (10 Hrs)	8085 Microprocessor & Instruction set: Introduction to microprocessor – Architecture– pin out diagram– addressing modes, Memory interfacing, Brief description of the instruction set – programming using data transfer group, arithmetic group, logical group, branch transfer group, stack and subroutines							
UNIT-II (10 Hrs)	8086 Microprocessor & Instruction set: Architecture and Programming of 8086 microprocessor: pipelining, Instruction sets, addressing modes – Memory addressing, decoding and Memory interfacing – Interrupts and interrupts handling.							

UNIT-III (10 Hrs)	Interfacing peripherals to 8086: Detailed architecture and configuration of ICs: 8251 USART, 8255 PPI, 8257 DMA Controller, 8253/8254 Programmable timer, 8279 programmable keyboard/display, A/D and D/A converters. Interfacing with 8086.
UNIT-IV (10 Hrs)	8051 Microcontroller: Introduction to 8051 Microcontroller, Architecture of 8051 Microcontroller, Memory Organization, Pin Diagram of 8051 Microcontroller, Timers/Counters, Serial Communication, Interrupts.
UNIT-V (10 Hrs)	Instruction Set, Programming and Applications of 8051 Microcontroller: Addressing Modes, 8051 Instruction Set, Simple Examples in Assembly-Language Programs of 8051 Microcontroller, interfacing programming for Traffic light controller, Stepper motor, washing machine using 8051 Microcontrollers.
Text Books:	
1.	Ramesh S. Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publishing, New Delhi, 5th edition, 2008.
2.	Advanced Microprocessors & Peripherals by A K Ray, K M Bhurchandi, Mc Graw hill publications
3.	Kenneth. J.Ayala, 8051 Microcontroller, Cengage learning, 3rd edition, 2005.
Reference Books:	
1.	Douglas V.Hall, "Microprocessors and Interfacing", 2nd Revised Edition, TMH Publications. Design", 2nd ed., PHI
2.	Krishna Kant, "Microprocessors and Microcontrollers", PHI Publisher

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3106	PE	3	--	--	3	25	75	3 Hrs.
ANALOG ELECTRONICS								
(For EEE)								
Course Objectives: Students will learn								
1.	Illustrate the analysis and design of feedback amplifiers							
2.	Illustrate the principle of oscillator and analyze different types of sinusoidal oscillators							
3.	To design linear and non-linear applications of op-amp							
4.	To analyze the concepts of Active filters and waveform generators							
5.	To design basic applications using 555 Timer							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Outline the concepts of feedback amplifiers							K2
2.	Analyse and design practical electronic circuits using amplifiers, oscillators							K4
3.	Design fundamental circuits using op-amps							K4
4.	Design and analyse of Active filters and oscillators							K4
5.	Design and analyse of various applications using 555 Timer and IC 565							K4
SYLLABUS								
UNIT-I (10 Hrs)	Feedback Amplifiers: Concept of Feedback Amplifiers - Effect of Negative Feedback on the amplifier characteristics. Four feedback topologies, Method of analysis of Voltage Series, Current Series, Voltage Shunt and Current Shunt feedback Amplifiers.							
UNIT-II (8 Hrs)	Sinusoidal Oscillators: Condition for oscillations and types of Oscillators – RC Oscillators: RC Phase Shift and Wien bridge Oscillators. LC Oscillators: Hartley, Colpitts, Clapp, Tuned Collector and Crystal Oscillators.							
UNIT-III (14 Hrs)	Applications of Operational Amplifiers: Basics of Op-Amp, Block Diagram, Ideal characteristics, op-amp parameters, open loop, and closed loop op-amp configurations, Applications of op-amp- summing amplifier, subtractor, Basic Integrator, Basic differentiator, Logarithmic amplifier; Instrumentation Amplifiers, Op-amp As a Comparators, Schmitt trigger, square wave generator, Rectifiers							
UNIT-IV (9 Hrs)	Active Filters and Oscillators: Butterworth type LPF, HPF first and second order filters, Higher order filters, Wide Band pass filters, Wide band reject filters, All pass filters. Op-Amp Phase Shift oscillator, Wein-bridge oscillator, Quadrature Oscillator, voltage-controlled oscillators.							

UNIT-V (8 Hrs)	Special ICs: 555 Timers, Introduction, Block diagram, 555 timer as an Astable and Monostable Multivibrator, Applications of 555 Timer as Monostable multivibrator-Frequency divider, Pulse stretcher, Pulse width modulation, Applications of 555 Timer as Astable Multivibrator- Square wave oscillator, FSK Generator, Schmitt trigger, IC 565 PLL – Block diagram, Monolithic IC565 PLL, Applications- Frequency multiplication/division, Frequency translation, Frequency adder/subtractor
Text Books:	
1.	Electronic Devices and Circuits by Salivahanan. Tata McGraw-Hill publication.
2.	Op-Amps and Linear ICs- RamakanthGayakwad, PHI, 1987.
Reference Books:	
1.	Integrated electronics-Jacob Millman and Christos C. Halkias
2.	Linear Integrated Circuits- D.Roy Chowdhury, New Age International(p) Ltd, 2nd Edition,2003.

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EE3107	PE	3	--	--	3	25	75	3 Hrs.	
LINEAR AND DIGITAL IC APPLICATIONS									
(For EEE)									
Course Objectives:									
1.	To design linear and non-linear applications of operational amplifier								
2.	To analyze the concepts of Active filters and waveform generators								
3.	To design the basic applications using 555 IC Timer, and IC 565 PLL								
4.	To analyze the concepts of Analog to digital and Digital to Analog converters								
5.	To analyze the concepts of various Digital Logic Families								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								K L
1.	Design/analyze fundamental circuits based on op-amps								K4
2.	Design and analyze of various active filters, oscillators								K4
3.	Design and Analyze of various applications using IC 555 timer and IC 565 PLL								K4
4.	Analyze the designing of Analog to digital and Digital to Analog converters								K4
5.	Analyze the various Digital IC Logic Families and to implement the logic function								k4
SYLLABUS									
UNIT-I (11 Hrs)	APPLICATIONS OF OPERATIONAL AMPLIFIERS: Basics of Op-Amp, Block Diagram, Ideal characteristics, op-amp parameters, open loop, and closed loop op-amp configurations, Applications of op-amp- summing amplifier, subtractor, Basic Integrator, Basic differentiator, Logarithmic amplifier; Instrumentation Amplifiers, Op-amp As a Comparators, Schmitt trigger, square wave generator (11)								
UNIT-II (9 Hrs)	ACTIVE FILTERS AND OSCILLATORS: Butterworth type LPF, HPF first and second order filters, Higher order filters, Wide Band pass filters, Wide band reject filter, All pass filters. Op-Amp Phase Shift oscillator, Wein-bridge oscillator, Quadrature Oscillator, Voltage to current and current to voltage converters. (9)								
UNIT-III (10 Hrs)	Special ICs 555, 565 & 566: Introduction, functional diagram of 555 IC, 555 timer as an Astable and Monostable Multivibrator, Applications of 555 Timer as Monostable multivibrator- Frequency divider, Pulse stretcher, Missing pulse detector, Linear ramp generator, Pulse width modulation, Applications of 555 Timer as Astable Multivibrator- Square wave oscillator, FSK Generator, Schmitt trigger. IC 565: Block diagram of PLL and applications of PLL IC 566 : Functional diagram of voltage controlled oscillator (IC 565)								
UNIT-IV (10 Hrs)	D to A and A to D CONVERTERS: Introduction, basic DAC techniques - weighted resistor DAC, R-2R ladder DAC, inverted R-2R DAC, A to D converters - parallel comparator type ADC, counter type ADC, successive approximation ADC and dual slope ADC.								

UNIT-V (12 Hrs)	Digital logic Families: Introduction, DTL, TTL, ECL, MOS logic, CMOS logic-analysis CMOS LOGIC: CMOS logic levels, MOS transistors, Basic CMOS Inverter, NAND and NOR gates, CMOS AND-OR-INVERT and OR-AND-INVERT gates. Applications of Digital ICs (TTL 74XX ICS): Study of logic gates using 74XX ICs,(IC 7483) Four-bit parallel adder , Comparator (IC 7485), Decoder (IC 74138, IC 74154), BCD-to-7- segment decoder (IC 7447), Encoder (IC 74147), Multiplexer (IC74151), Demultiplexer (IC 74154), BCD Counter 74LS90 BCD Counter
Text Books:	
1.	D. Roy Choudhury, Shail B. Jain (2012), Linear Integrated Circuit, 4th edition, New Age International Pvt.Ltd.,NewDelhi,India.
2.	Digital Design: Principles and Practices, 4th Edition John F. Wakerly,Pearson Publications.
Reference Books:	
1.	Ramakant A. Gayakwad, (2012), OP-AMP and Linear Integrated Circuits, 4th edition, Prentice Hall / Pearson Education, NewDelhi.
2.	Modern Digital Electronics, R.P.Jain, Fourth Edition
3.	Pulse and Digital Circuits, A. Anand Kumar, PHI publications
4.	Floyd, Jain (2009), Digital Fundamentals, 8th edition, Pearson Education, New Delhi.
e-Resources:	
1.	https://www.youtube.com/watch?v=ryYh89GpPy0&t=5s
2.	https://www.youtube.com/watch?v=yt1Fwrc8ysk&t=105s
3.	https://www.youtube.com/watch?v=9Y8zyga5Hcs
4.	https://www.youtube.com/watch?v=Gy18D6HSFaM&t=58s

Code	Category	L	T	P	C	I.M	E.M	Exam
R19EE3108	PE	3	--	--	3	25	75	3 Hrs.
PULSE AND DIGITAL CIRCUITS								
Course Objectives:								
1.	To provide insight of the applications of Integrator, differentiator circuits.							
2.	To introduce the design of various clippers circuits and to provide insight of the applications of clamper circuits.							
3.	To introduce the analysis of Bistable, Monostable, Astable Multivibrators and Schmitt-trigger for various applications.							
4.	To introduce various Time Base Generators.							
5.	To provide insight of the synchronization techniques for sweep circuits and to provide insight of different logic families; realize logic gates using diodes and transistors.							
Course Outcomes:								
S.No	Outcome							Knowledge Level
1.	Analyze passive RC circuits and understand the applications of Integrator, differentiator circuits.							K4
2.	Design of different clipping circuits and understand the applications of clamper circuits.							K4
3.	Design different multivibrators for various applications.							K4
4.	Understand different Time Base Generators.							K2
5.	Analyze synchronization techniques for sweep circuits and understand different logic families; realize logic gates using diodes and transistors.							K4
SYLLABUS								
UNIT-I (10 Hrs)	LINEAR WAVE SHAPING: High pass, Low pass RC circuits, Their response for sinusoidal, step, pulse, square and ramp inputs, The High pass RC circuit as a differentiator and the Low pass RC circuit as an integrator, Attenuators.							
UNIT-II (10 Hrs)	NONLINEAR WAVE SHAPING: Diode clippers, Clippers at two independent levels, Transistor clipper, Emitter coupled clipper, Comparators, Applications of voltage comparators, clamping operation, clamping circuits taking source and diode resistances into account - transient and steady state response for a square wave input, Clamping circuit theorem, Design of a clamping circuit.							
UNIT-III (12 Hrs)	MULTIVIBRATORS: Bi-stable multivibrators: Transistor as a Switch, Transistor Switching times, Bistable Multivibrator-explanation, A Fixed-bias and A Self-biased transistor binary, Commutating capacitors-analysis, Triggering the binary, and Schmitt trigger circuit-emitter coupled binary circuit. Mono-stable multivibrator: Design and analysis of Collector coupled Monostable							

	Multivibrator, Expression for the gate width and its waveforms Astable multivibrator: Design and analysis of Collector coupled Astable Multivibrator, Expression for the Time period and its waveforms, The Astable Multivibrator as a voltage to frequency converter.
UNIT-IV (10 Hrs)	TIME –BASE GENERATORS: General features of a Time base signal, Methods of generating a time base waveform, Exponential sweep circuit - Errors that define Deviation from linearity, UJT Relaxation Oscillator, Miller and Bootstrap Time base generators, Current Sweep generator, Linearity correction through adjustment of driving Waveform.
UNIT-V (10 Hrs)	SYNCHRONIZATION AND LOGIC FAMILIES: Synchronization and frequency division: Pulse synchronization of relaxation devices, frequency division in the sweep circuit, Synchronization of Astable multivibrator, Synchronization of a sweep circuit with symmetrical signals. Digital logic Families: Introduction, RTL, DTL, TTL, ECL, MOS logic, CMOS logic-analysis
Text Books:	
1.	Pulse, Digital and switching wave forms by Milliman and Taub Mc Graw Hill.
2.	Pulse and Digital Circuits by A. Anand Kumar, PHI.
Reference Books:	
1.	Pulse and Digital Circuits by MS Prakash Rao, Tata McGraw Hill.
2.	Pulse and Digital Circuits by Venkatrao K., Ramasudha K., Manmadharao. G, Pearson education, 2010.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3109	PE	--	--	--	3	25	75	3 Hrs.
MOOCS-I								
(For EEE)								
MOOCs-I course should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3110	PC	--	--	3	1.5	20	30	3 Hrs.
ELECTRICAL MACHINES- I LAB								
(For EEE)								
Course Objectives: Students will Conduct								
1.	Load test on DC shunt and DC series motors and Hopkinson’s test.							
2.	OC and SC tests and load test on single phase transformer.							
3.	Swinburn’s test and sumpner’s test on DC shunt motor and transformer							
4.	OCC of DC shunt generator and load test on DC compound generator							
5.	Speed control of DC shunt motors.							
Course Outcomes: Students will be able to								
S.No	Outcome							Knowledge Level
1.	Conduct Loadtests on DC motors and Hopkinson’s test toAnalyse the efficiency performance of DC Shunt and DC series motors.							K4, K5
2.	ConductOC and SC tests and Load test on transformer toAnalyse the regulation and efficiency performance of the transformer.							K4, K5
3.	ConductSwinburn’s test and Sumpner’s test to Predict the Efficiency performance of Dc shunt machine and Transformer.							K4, K5
4.	ConductOCCDC Shunt generator and Load test on DC compound generator to Determine the open circuit characteristics and over compound and under compound characteristics.							K4, K5
5.	Conduct a test onSpeed control of a DC shunt motor to Analyse the Speed characteristics of DC shunt motor.							K4, K5
LIST OF EXPERIMENTS								
1.	Swinburn’s Test							
2.	Speed control of a DC shunt motor.							
3.	Load test on DC Shunt motor.							
4.	Load test on DC series motor.							
5.	Load test on DC Compound generator.							
6.	Open circuit characteristics of a DC shunt generator.							
7.	Hopkinson's Test.							
8.	Internal and external characteristics of a DC shunt generator.							
9.	OC and SC tests on a single-phase transformer.							
10.	Load test on a single-phase transformer.							
11.	Sumpner's Test.							
Reference Books:								
1.	Electrical Machine by Dr. P. S. Bhimbra Khanna publications, 2011.							

2.	Irving L. Kosow, “Electrical Machines & Transformers”, Prentice Hall; 2 nd Revised Edition 1990.
3.	Clayton, A.E; “Performance and Design of direct machines” CBS; IST edition (2004).

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3111	PC	--	--	3	1.5	20	30	3 Hrs.
MICROPROCESSOR & MICROCONTROLLER LAB								
(For EEE)								
Course Objectives: Student will								
1.	Learn about working of 8085, 8086 microprocessor and instructions sets.							
2.	Acquire knowledge of writing program for arithmetic, logical operation to perform required task like pick largest even number, 8-bit array addition etc.							
3.	Understand the programs for converting binary to BCD codes and also learns about sorting techniques.							
4.	Learn about working of 8051 microcontroller and instructions sets							
5.	Learn about basic programming skills using 8051 controller and also acquire knowledge on interfacing with other devices like traffic light controller, stepper motor etc							
Course Outcomes: Students will be able to								
S.No	Outcome							KL
1.	Demonstrate the architecture, instruction set, configuration of 8085 ,8086 processors and 8051 controllers							K3
2.	Develop simple arithmetic and logical programs on 8085 and 8086 processors using the instruction set.							K5
3.	Evaluate programs on sorting, counting, Binary to BCD conversion on 8085 & 8086 processors using the instruction set.							K5
4.	Develop arithmetic and logical programs on 8051controller using the instruction set.							K5
5.	Design programs for interfacing circuits like traffic controller, LED display board, Motor controllers etc.							K5
LIST OF PROGRAMS								
PART A: PROGRAMS ON MICROPROCESSORS:								
1.	Program to add two 8-bit binary numbers on 8085 processor							
2.	Program to add an array of 8-bit binary numbers on 8085 processor.							
3.	Program to pick the largest even number from an array of 8- bit binary numbers on 8085 processor							
4.	Program to find the sum of an array of 2- digit packed BCD numbers on 8085 processor.							
5.	Program to convert an 8- bit binary number into BCD on 8085 processor.							
6.	Program to sort given array of 8-bit binary numbers on 8085 processor.							
7.	Program for 16-bit arithmetic operations on 8086 micro-processor (using Various Addressing Modes).							
8.	Program for sorting a given array using 8086 micro-processor							

PART B: PROGRAMS ON MICRO CONTROLLERS:	
1.	Microcontroller programming on two 8-bit numbers a) Addition, b) Subtraction, c) Multiplication & d) Division
2.	Program to obtain decimal equivalent of an 8-bit hexadecimal number
3.	Program to find the sum of array of 8-bit binary numbers
4.	Interfacing stepper motor and speed control using 8051 microcontrollers
5.	Traffic light control with 8051 microcontrollers.
Text Books:	
1.	Ramesh S. Gaonkar, Microprocessor Architecture, Programming, and Applications with the 8085, Penram International Publishing, New Delhi, 5th edition, 2008.
2.	Advanced Microprocessors & Peripherals by A K Ray, K M Bhurchandi, Mc Graw hill publications
3.	Kenneth. J.Ayala, 8051 Microcontroller, Cengage learning, 3rd edition, 2005.
Reference Books:	
1.	Douglas V.Hall, “Microprocessors and Interfacing”, 2nd Revised Edition, TMH Publications.Design”, 2nd ed., PHI

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3101	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS - I								
Common to All Branches								
Part-A: Verbal and Soft Skills-I								
Course Objectives:								
1.	To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.							
2.	To familiarize the learner with high frequency words as they would be used in their professional career.							
3.	To inculcate logical thinking in order to frame and use data as per the requirement							
4.	To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.							
5.	To familiarize students with soft skills and how it influences their professional growth.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.							K3
2	Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.							K3
3.	Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.							K3
4.	Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.							K3
5.	Apply soft skills in the work place and build better personal and professional relationships making informed decisions.							K3
SYLLABUS								
UNIT-I	Grammar: (VA) Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns- number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.							
UNIT-II	Vocabulary: (VA) Synonyms and synonym variants (with emphasis on high frequency words), antonyms and antonym variants (with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple							

	meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.
UNIT-III	Reasoning: (VA) Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).
UNIT-IV	Usage: (VA) Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.
UNIT-V	Soft Skills: Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.
Text Books:	
1.	Oxford Learners,,s Grammar – Finder by John Eastwood, Oxford Publication.
2.	R S Agarwal,,s books on objective English and verbal reasoning
3.	English Vocabulary in Use- Advanced , Cambridge University Press
4.	Collocations In Use, Cambridge University Press
5.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi
Reference Books:	
1.	English Grammar in Use by Raymond Murphy, CUP
2.	Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	The Art of Public Speaking by Dale Carnegie
5.	The Leader in You by Dale Carnegie
6.	Emotional Intelligence by Daniel Golman
7.	Stay Hungry Stay Foolish by Rashmi Bansal
8.	I have a Dream by Rashmi Bansal.
Part-B: Quantitative Aptitude-I	
Course Objectives:	
1.	To familiarize students with basic problems on numbers and ratios problems.

2.	To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.	
3.	To enable the students to work efficiently on percentage values related to shares, profit and loss problems.	
4.	To inculcate logical thinking by exposing the students to reasoning related questions.	
5.	To inculcate logical thinking by exposing the students to reasoning related questions.	
Course Outcomes:		
S.No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating on number problems and various units of ratio concepts	K3
2.	Accurate solving problems on time and distance and units related solutions	K3
3.	The students will become adept in solving problems related to profit and loss, in specific, quantitative ability	K3
4.	The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry	K3
5.	The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies	K3
SYLLABUS		
UNIT-I	Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion	
UNIT-II	Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.	
UNIT-III	Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends. Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.	
UNIT-IV	Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter	

	series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.
UNIT-V	Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal
3.	Puzzles to puzzle you by shakunataladevi.
References:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for cat by arunsharma.
5.	Elementary and Higher algebra by HS Hall and SR knight.
Websites:	
1.	www.m4maths.com
2.	www.Indiabix.com
3.	www.800score.com
4.	Official GRE site
5.	Official GMAT site

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3102	MC	--	--	3	--	--	--	--
BASIC CODING								
(Common to ECE and EEE)								
Course Objectives:								
1.	To develop programming skills among the students							
2.	To familiarize the student with Control Structures, Loop Structures							
3.	To familiarize the student with Basic searching and sorting Methods							
4.	To familiarize the student with Functions, Recursions and Storage Classes							
5.	To familiarize the student with Structures and Unions							
6.	. To familiarize the student with Operating System concepts.							
7.	To familiarize the student with Networking concepts.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Know about Control Structures, Loop Structures and branching in programming							K3
2	Know about various searching and sorting methods.							K3
3.	Know about Functions, Recursions and Storage Classes							K3
4.	Know about Structures and Unions.							K3
5.	Know different Operating System concepts. 6.							K3
6.	Differentiate OSI Model Vs. TCP/IP suite.							K3
SYLLABUS								
UNIT-I								
Review of Programming constructs Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.								
UNIT-II								
Introduction to Linear Data, strings and pointers Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.								
UNIT-III								
Functions, Recursions and Storage Classes Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions. Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime- memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.								

UNIT-IV	User-defined data types, Pre-processor Directives and standard storage Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming. Practice: Structure padding, user-defined data storage and retrieval programs
UNIT-V	Operating system principles and Database concepts: Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer.
Reference Books:	
1.	Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2.	The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd.
3.	Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), EnzoMarinari (Author), Giovanni Organtini, World Scientific
4.	Object Oriented Programming in C++: N. Barkakati, PHI.
5.	Object Oriented Programming through C++ by Robat Laphore.
6.	https://www.geeksforgeeks.org/ .
7.	https://www.tutorialspoint.com/

ELECTRICAL & ELECTRONICS ENGINEERING

(Accredited by NBA)

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R19)

III/IV B.TECH**II-SEMESTER**(With effect from **2019-2020** Admitted Batch onwards)

Subject Code	Name of the Subject	Category	Cr	L	T	P	Internal Marks	External Marks	Total Marks
B19 EE 3201	Power System Analysis and Stability	PC	3	3	--	--	25	75	100
B19 EE 3202	Internet of Things and its applications in Electrical Engineering	PC	3	3	--	--	25	75	100
# PE-II	Professional Elective-II	PE	3	3	--	--	25	75	100
# PE-III	Professional Elective-III	PE	3	3	--	--	25	75	100
# OE-I	Open Elective-I	OE	3	3	--	--	25	75	100
B19 EE 3211	Electrical Machines - II Lab	PC	1.5	--	--	3	20	30	50
B19 EE 3212	Control Systems Lab	PC	1.5	--	--	3	20	30	50
B19 EE 3213	Power Electronics Lab	PC	1.5	--	--	3	20	30	50
B19 MC 3201	Employability skills-II	MC	0	3	--	--	--	--	--
B19 MC 3203	Advanced Coding	MC	--	--	--	3	--	--	--
TOTAL			19.5	18	--	12	185	465	650

#PE-II	Course Code	Course
	B19 EE 3203	Advanced Control Systems
#PE-III	B19 EE 3204	Digital Control systems
	B19 EE 3205	Special Electrical Machines
	B19 EE 3206	MOOCs-II
	B19 EE 3207	Power Electronic Drives
	B19 EE 3208	Advanced power electronics Circuits
	B19 EE 3209	Switched Mode Power Supplies
	B19 EE 3210	MOOCs-III

	Course Code	Course
#OE-I	Student has to study one Open Elective offered by CE or CSE or ECE or IT or ME or EM&H	

Code	Category	L	T	P	C	I.M	E.M	Exam	
B19EE3201	PC	3	--	--	3	25	75	3 Hrs.	
POWER SYSTEM ANALYSIS AND STABILITY									
(For EEE)									
Course Objectives:									
1.	Single line diagram, per unit system, per unit reactance diagram.								
2.	Formation of Y-bus, Gauss-seidel method, Newton-Raphson method, fast decoupled method.								
3.	Short circuit momentary current, interrupting current and short circuit MVA.								
4.	Symmetrical components, sequence networks, LG, LL and LLG Faults on a power system.								
5.	Swing equation, equal area criterion, steady state stability and transient stability of a power system.								
Course Outcomes: At the end of this course, the students will be able to									
S.No	Outcome								K L
1.	Compute pu reactance and pu impedance and draw per unit reactance diagram.								K3
2.	Apply load flow techniques to analyze load flow problems in the power system.								K3, K4
3.	Compute short circuit MVA and analyze symmetrical fault currents and voltages in a power system.								K3, K4
4.	Determine symmetrical components and draw sequence networks of power system components and calculate un-symmetrical fault currents in a power system.								K3
5.	Apply swing equation and equal area criterion to analyze steady state and transient stability problems of a power system.								K3, K4
SYLLABUS									
UNIT-I (10 Hrs)	P.U. REPRESENTATION: Single Line Diagram, Per Unit Quantities, P.U. Impedance of 3-Winding Transformers, P.U. Impedance Diagram of a Power System.								
UNIT-II (9 Hrs)	LOAD FLOW STUDIES: Formulation of Network Model Formulation, Load Flow Problem, Gauss-Seidel Method, Newton-Raphson Method & Fast Decoupled Method of Solving Load Flow Problem.								
UNIT-III (10 Hrs)	SYMMETRICAL FAULT ANALYSIS: Transient on a Transmission line, Short Circuit of Synchronous Machine (No Load), Short Circuit of a loaded Synchronous Machine, Short Circuit MVA Calculations, Fault Limiting Reactors.								
UNIT-IV (10 Hrs)	SYMMETRICAL COMPONENTS and SEQUENCE NETWORKS: The Symmetrical Components, Sequence Impedances and Sequence Networks, Phase Shift in Star/ Delta Transformers, 3-Phase Power in terms of Symmetrical Components. UN-SYMMETRICAL FAULTS: LG, LL and LLG Faults on a power system under Unloaded Condition.								
UNIT-V (12 Hrs)	POWER SYSTEM STABILITY: Dynamics of Synchronous Machine, Swing Equation, Steady State Stability, Transient Stability, Equal Area Criterion, Critical Clearing Angle and Critical Clearing time, Solution of Swing Equation by Step-by-Step Method, Factors Affecting Transient Stability.								

Text Books:	
1.	Power System Analysis by D.P. Kothari & I.J. Nagarath, 4th Edition, TMH Publication.
2.	Elements of Power System Analysis, William D. Stevenson, Jr, Mc Graw Hill Pub first edition, 1955.
Reference Books:	
1.	Power System Analysis by Hadi Sadat, Mc Graw Hill Pub 1999.
2.	Electrical Power Systems by C.L. Wadhwa, 7th Edition, New Age International Publications, 2017.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3202	PC	3	--	--	3	25	75	3 Hrs.
INTERNET OF THINGS AND ITS APPLICATIONS IN ELECTRICAL ENGINEERING								
(For EEE)								
Course Objectives:								
1.	To study need of IoT, challenges while using IoT devices in various application areas							
2.	To study the various sensors used in IoT Networks, Raspberry Pi, Arduino and their peripherals,							
3.	To study the power management in IoT networks							
4.	To study the wireless communication devices and their communication protocols used in IoT networks							
5.	To study development of IoT application from a global perspective and building automation systems by considering real world constraints.							
Course Outcomes: At the end of this course, the students will be able to								
S.No	Outcome							K L
1.	Identity different components of IoT Networks							K3
2.	Interface various sensors to Processor boards							K3
3.	Design Power management circuit for IoT application							K3
4.	Apply various IoT network protocols to communicate between sensors and machines wirelessly							K3
5.	Develop automation of a system using IoT devices							K3
SYLLABUS								
UNIT-I (8 Hrs)	Introduction to IoT: What is IoT, Industrial IoT, Role of M2M in IoT , Cyber Physical Systems, Elements of an IoT ecosystem–various devices, Sensing, Actuation;, Basics of Networking –OSI model ,IP.V4,IP.V6;Challenges in design IoT networks							
UNIT-II (12 Hrs)	Processor boards &Sensor Interface Arduino board and its Peripherals, Raspberry Pi and its Peripherals, Sensors- Sensor Network, Sensor-Cloud, interfacing;							
UNIT-III (10 Hrs)	Power Management in IoT Networks: Need of power management in IoT networks, Power supply systems used for IoT networks- Low dropout regulator, switching regulator, Buck Boost Converters, Linear regulator; Energy harvesting - PV, RF, Kinetic Energy, TEGs,							
UNIT-IV (12 Hrs)	Data Communication & Protocols: Wireless Interfaces - LAN - BLE, Wi-Fi, Zigbee, RFID, LP WAN –(LORA, LTE-M, NB-IOT), Protocols –(REST based COAP, MQTT, AMQP, SDN for IoT, Cloud Connectivity in IoT networks, CAN)							

UNIT-V (8 Hrs)	IoT Applications IoT in Power Plants, Smart Grid, Smart energy monitoring, Electrical Vehicles, Connected Vehicles.
Text Books:	
1.	Vijay Madiseti and ArshdeepBahga, “Internet of Things (A Hands-on-Approach)”, 1 stEdition,VPT, 2014
2.	Simon Monk, “Programming the Raspberry Pi: Getting Started with Python”, January 2012, McGraw Hill Professional Massimo Banzi, “Getting Started with Arduino”, First Edition, February 2009, O'Reilly Media, Inc
3.	Olivier Hersent, David Boswarthick, Omar Elloumi , “The Internet of Things – Key Applications and Protocols”, Wiley, 2012.
Reference Books:	
1.	Adrian McEwen, Hakim Cassimally, “Designing the Internet of Things”, November 2013, John Wiley and Sons.
2.	Holger Kerl, Andreas Willig, “Protocols and Architectures for Wireless Sensor Network”, John Wiley and Sons, 2005 (ISBN: 978-0-470-09511-9)

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3203	PE	3	--	--	3	25	75	3 Hrs.
ADVANCED CONTROL SYSTEMS								
(For EEE)								
Course Objectives: Students will learn								
1.	About various components used in practical control systems.							
2.	The design of basic compensators using Root-locus and frequency response.							
3.	To model physical systems using state space approach and verify the system response.							
4.	The principles of state feedback controllers.							
5.	Understand how a digital computer can be used in control systems.							
Course Outcomes: Students will be able to								
S. No	Outcome							Knowledge Level
1.	Employ various components in the implementation of control systems.							K3
2.	Apply root locus and Bode plot techniques for designing basic compensators.							K3, K4
3.	Construct state space models for control systems and analyze them.							K3
4.	Analyze control systems with state feedback.							K4
5.	Apply digital computers for control system applications							K3
SYLLABUS								
UNIT-I (9 Hrs)	CONTROL SYSTEMS COMPONENTS: Position sensors: Potentiometers, Optical rotary encoders, Linear variable differential transformers (LVDTs)-D.C. & A.C. Tachometers-A.C. and D.C. Servo Motors -Synchro's-Stepper Motors and their use in Control Systems, Other sensors for temperature, pressure, flow etc.							
UNIT-II (11 Hrs)	CLASSICAL DESIGN OF CONTROL SYSTEMS: Introduction to Classical Design-Cascade compensation design using Root-locus-Ideal Integral Compensation (PI controller)-Lag Compensator-Ideal Derivative Compensation (PD controller)-Lead Compensator-PID controller- Lag-lead-compensator -Cascade Compensation design in Frequency Domain (Bode Plot Techniques) - Realization of Compensating Networks							
UNIT-III (10 Hrs)	STATE VARIABLE ANALYSIS: Concept of State and State Variables, State Models for Linear Continuous Time Systems, Different forms of state space models (phase variable, diagonal, CCF, OCF), Conversion between state space and transfer function models, Solution of State Equations: Laplace transform approach and Time domain approach -State Transition Matrix.							
UNIT-IV (9 Hrs)	DESIGN OF STATE FEEDBACK CONTROL SYSTEMS: Introduction to state feedback control, Pole Placement by State Feed-Back, Design of Controllers for pole placement, Concept of Controllability and Observability, Steady-State Error Design via Integral Control.							

UNIT-V (11 Hrs)	DIGITAL CONTROL SYSTEMS: Introduction to Digital control systems, Discrete time systems, Sampled Data Systems, z-domain transfer functions (Pulse transfer functions), Analysis of digital control systems, Stability analysis in Z domain, Jury's test, Bi-linear transformation.
Text Books:	
1.	I. J. Nagrath and M. Gopal, 'Control Systems Engineering', New Age International Publishers (6th Edition).
2.	Norman S. Nise, 'Control Systems Engineering', Wiley publications (8th Edition)
Reference Books:	
1.	Richard C. Dorf and Robert H. Bishop, 'Modern Control Systems', Addison-Wesley Publishers (8th Edition)
2.	Kilian, 'Modern Control Technology: components and Systems', 2 nd edition, online

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3204	PE	3	--	--	3	25	75	3 Hrs.
DIGITAL CONTROL SYSTEMS								
(For EEE)								
Course Objectives: Student will learn								
1.	To understand the significance of signal conversion by using S/H device and characteristic of ZOH& FOH.							
2.	To Learn about Z-transforms and relation between Laplace & Z-transform.							
3.	To study the Pulse transfer functions for the systems incorporated with Sample and Hold devices.							
4.	To Learn the concept of state variable technique for discrete systems.							
5.	To Learn the concept of stability and tests used in digital control systems.							
Course Outcomes: Student will be able to								
S.No	Outcome							Knowledge Level
1.	Apply the concepts of Signal conversion to examine the characteristics of ZOH.							K3
2.	Apply Z- transform in Engineering application related to digital control systems							K3
3.	Determine the pulse transfer function in digital systems which is useful to relate input and output of a given system.							K3
4.	Apply the concept of State variables to analyze multi- input- multi- output systems of linear time invariant systems.							K3, K4
5.	Apply the concept of Controllability and Observability to analyze the behavior of state variables and test the stability of linear digital system by Bi-Linear Transformation Method- Jury's test.							K3, K4
SYLLABUS								
UNIT-I (10 Hrs)	SIGNAL CONVERSION AND PROCESSING Introduction, Block diagram representation of Sample and Hold Device, Mathematical Modelling of the Sampling Process, Finite -Pulse Width Sampler, The Sampling theorem, Mathematical Modelling of The Sampling, Ideal Sampler, Sample and Hold Devices, Expressions of F*(S), S-Plane Properties of F*(S), Zero-Order Hold, Frequency-Domain Characteristics of ZOH, First Order Hold, Fractional Hold Device.							
UNIT-II (10 Hrs)	Z- TRANSFORM Definition, Relationship with Laplace Transform, Alternate Expression for F(Z), Evaluation of Z-Transform, Relationship Between S-Plane And Z-Plane, Inverse Z-Transform, Non-Uniqueness of Z-Transform, Defining Equations of the inverse Z-Transform, Theorems of Z-Transform.							
UNIT-III (08 Hrs)	TRANSFER FUNCTIONS AND BLOCK DIAGRAMS Transfer Functions, Block diagrams, The Pulse Transfer Function And Z-Transform Function, Systems with Cascaded Elements Separated by a Sampler & Not Separated by a Sampler, Pulse Transfer Function of ZOH and Relation Between G(S) And G(Z), Closed Loop Systems, Characteristic Equation- problems.							

UNIT-IV (12 Hrs)	STATE VARIABLE TECHNIQUES State equation and state transition equations of continuous data systems. State transition matrix solutions, properties of state transition matrix, solution of non-homogeneous state equations, state equations of discrete systems with sample and hold devices, state transition equations, the recursive method, the z-transform method, state equations and transfer function, characteristic Equation, Eigen Values, Eigen Vectors, Diagonalization of the 'A' Matrix, Jordan Canonical Form-Methods for Computing State Transition Matrix-problems.
UNIT-V (10 Hrs)	CONTROLLABILITY, OBSERVABILITY AND STABILITY Definition of Controllability and Observability, Theorem on Controllability and Observability, Relationships between Controllability and Observability and Transfer Function, Stability of Linear Digital Control Systems, Stability Tests-Bi-Linear Transformation Method- Jury's Stability test-problems.
Text Books:	
1.	Digital Control Systems, B. C. KUO, Oxford University Press, 2 nd Edition
Reference Books:	
1.	Discrete time control systems, K. Ogata, Pearson Education/PHI, 2 nd Edition
2.	Digital control and state variable methods by M. Gopal, Tata McGraw Hill.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3205	PE	3	--	--	3	25	75	3 Hrs.
SPECIAL ELETRICAL MACHINES								
(For EEE)								
Course Objectives: Students will								
1.	Study the construction, principle of operation and control of Stepper Motor.							
2.	Study the construction, principle of operation and control of Permanent Magnet Brushless DC.							
3.	Study the construction, principle of operation control of Permanent Magnet Synchronous Motors.							
4.	Study the construction, principle of operation of Switched Reluctance Motor.							
5.	Study the construction, principle of operation of Synchronous Reluctance Motor.							
Course Outcomes: Student will be able to								
S.No	Outcome							Knowledge Level
1.	Illustrate the principle of operation and control for stepper motor.							K3
2.	Acquire the knowledge on principle of operation, and control of Permanent Magnet Brushless DC Motor							K3
3.	Illustrate the operation and control of Permanent Magnet Synchronous motor.							K3
4.	Acquire the knowledge on principle of operation and control of Switched reluctance Motor.							K3
5.	Illustrate the operation and control of Synchronous reluctance Motor.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Stepper Motor: Introduction, Variable Reluctance Stepper Motor, Permanent Magnet Stepper Motor, Hybrid Stepper Motor, Torque Equation, Characteristics of Stepper Motor, Open & Closed – loop Control of Stepper Motor, Applications of Stepper Motors							
UNIT-II (10 Hrs)	Permanent Magnet Brushless DC Motor: Permanent Magnet materials –Principle of operation – Types –EMF and torque equations – Commutation - Power Converter Circuits and their controllers – Motor characteristics and control– Applications.							
UNIT-III (8 Hrs)	Permanent Magnet Synchronous Motor (PMSM): Construction, Principle of Operation, EMF Equation, Torque Equation, Phasor Diagram, Comparison for Conventional Synchronous Motor and PMSM, Control of PMSM, Applications.							
UNIT-IV (8 Hrs)	Switched Reluctance Motor (SRM): Constructional features, Principle of Working, Basics of SRM Analysis, Torque Equation and Characteristics, Power Converter Circuits & Control of SRM, Rotor Position Measurement and Applications.							

UNIT-V (10 Hrs)	Synchronous Reluctance Motor (SYRM): Constructional features – Types – Axial and Radial flux motors – Working principle – Variable Reluctance Motors – Voltage and Torque Equations - Phasor diagram - performance characteristics – Applications.
Text Books:	
1.	T. Kenjo, ‘Stepping Motors and Their Microprocessor Controls’, Clarendon Press London, 1984
2.	T.J.E. Miller, ‘Brushless Permanent Magnet and Reluctance Motor Drives’, Clarendon Press, Oxford, 1989
3.	K. Venkataratnam, ‘Special Electrical Machines’, Universities Press (India) Private Limited, 2008
Reference Books:	
1.	R.Krishnan, ‘Switched Reluctance Motor Drives – Modeling, Simulation, Analysis, Design and Application’, CRC Press, New York, 2001
2.	T. Kenjo and S. Nagamori, ‘Permanent Magnet and Brushless DC Motors’, Clarendon Press, London, 1988
3.	E.G. Janardanan, ‘Special electrical machines’, PHI learning Private Limited, Delhi, 2014

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3206	PE	3	--	--	3	25	75	3 Hrs.
MOOCS-I								
(For EEE)								
MOOCs-I course should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.								

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3207	PE	3	--	--	3	25	75	3 Hrs.
POWER ELECTRONIC DRIVES								
(For EEE)								
Course Objectives: Students will learn								
1.	Classifications and characteristics of Drive and choosing the suitable drive for various applications.							
2.	Dynamics of Drive with mechanical characteristics.							
3.	Analyze rectifier fed DC Drives.							
4.	Analyze chopper fed DC Drives.							
5.	Analyze various types of Induction motor Drive with slip power recovery schemes.							
Course Outcomes: Students will be able to								
S.No.	Outcomes							Knowledge level
1.	Select drive for particular application & demonstrate the operation of drive to satisfy four-quadrant operation to meet load requirements.							K3
2.	Analyze the dynamics of electric drive & also Explain starting and braking methods.							K3, K4
3.	Apply and analyze rectifier fed DC drives for continuous and discontinuous modes.							K3, K4
4.	Apply and analyze chopper fed DC drives for continuous and discontinuous modes with closed loop control.							K3, K4
5.	Apply control strategies for Induction motor Drives and also distinguish slip power recovery schemes.							K3, K4
SYLLABUS								
UNIT-I (9 Hrs)		Introduction to Electric Drives: Basics of Power Electronic Drive system and components- (Sources, Electric Motors, Power Modulators), Advantages of Electric Drives, Choice of Electric Drives and Selection of Drive for various applications, Multi-Quadrant operation of the drive.						
UNIT-II (10 Hrs)		Dynamics of Electric Drive: Fundamental torque equation of electric drive, Equivalent values of drive parameters, Measurement of the moment of inertia, Different types of loads, Stability of power electronic drives, Load equalization, Different electric Starting and Braking methods.						
UNIT-III (10 Hrs)		Rectifier Controlled Fed DC Drives: Single Phase Fully controlled converters connected to DC separately excited motor and DC series motor – Continuous & Discontinuous current operation – voltage and current waveforms – Speed Torque expressions – Speed Torque Characteristics.						

UNIT-IV (10 Hrs)	Chopper Controlled Fed DC Drives: Chopper controlled DC separately excited motor and DC series motor – Continuous current operation – voltage and current waveforms – Speed Torque expressions – Speed Torque characteristics, Closed-loop control of DC drive.
UNIT-V (11 Hrs)	Control of Induction Motors: Variable voltage control of Induction motor by AC voltage controller, Variable frequency control of Induction motor by cycloconverter – waveforms – Speed Torque characteristics, V/f Control Method, Slip power recovery schemes – Static Kramer Drive – Static Scherbius Drive.
Text Books:	
1.	Fundamentals of Electrical Drives by G.K.Dubey, Second Edition, 2002.
2.	Power Electronics: Circuits, Devices and Applications by M.H.Rashid, Third Edition, 2009.
Reference Books:	
1.	Modern Power Electronics and AC Drives by Bimal K Bose, 2005.
2.	Electric motor drives: modeling, analysis, and control by R Krishnan, 2001.
3.	Thyristor Control of Electric Drives by Vedam Subramanyam, Tata McGraw-Hill Publications, 2008.

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3208	PE	3	--	--	3	25	75	3 Hrs.
ADVANCED POWER ELECTRONIC CIRCUITS								
(For EEE)								
Course Objectives: student will learn								
1.	To understand the necessity of AC-AC converters and their applications							
2.	To identify the requirement of power factor correction converters.							
3.	To acquaint basics of isolated DC-DC converters							
4	To distinguish various PWM techniques used in inverters.							
5.	To familiarize multi-level inverter configuration to improve the quality of the inverter output voltage.							
Course Outcomes:								
S.No	Student will be able to							Knowledge Level
1.	Analyze the RMS output voltage of three phase AC-AC Converters for R and RL loads.							K3, K4
2.	Identify the necessity of power factor correction converters.							K3
3.	Compare the operation of flyback, push pull and bridge type isolated DC-DC converters.							K4
4	Examine the voltage control of inverters using various PWM methods							K4
5.	Distinguish between flying capacitor, diode clamped and cascaded type multi-level inverters.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Three Phase AC Voltage Controllers Three phase AC Voltage controller Circuit operation Analysis with star and delta connected resistive, resistive-inductive loads -Effects of source and load inductances-numerical problems.							
UNIT-II (10 Hrs)	Power Factor Correction Converters: Single-phase single stage boost power factor corrected rectifier, power circuit principle of operation, and steady state- analysis, three phase boost PFC converter.							
UNIT-III (10 Hrs)	Isolated DC-DC Converters: fly-back converter, push-pull converter, Half bridge and full bridge converters circuit operation analysis and numerical problems.							
UNIT-IV (10 Hrs)	PWM Inverters: Principle of operation-Voltage control of single-phase inverters - sinusoidal PWM – modified PWM – phase displacement Control – Trapezoidal, staircase, stepped, harmonic injection and delta modulation. Voltage Control of Three-Phase Inverters- Sinusoidal PWM- 60° PWM- Third Harmonic PWM							

UNIT-V (10 Hrs)	Multi-level inverters: Types of Multi-level Inverters- Diode-Clamped Multilevel Inverter, Flying-Capacitors, Cascaded Multilevel Inverter-Comparisons of Multilevel Converters.
Text Books:	
1.	Power Electronics-Md.H.Rashid –Pearson Education Third Edition- First Indian Reprint- 2008.
2.	Ned Mohan, T.M. Undeland, William P.Robbins: Power Electronics: Converters, Applications, 3 rd ed., John Wiley & Sons, 2009.
Reference Books:	
1.	Pulse Width Modulation: Analysis and Performance in Multi-level Inverters, Satish Kumar Peddapelli.Walterde Gruyter GmbH, Berlin/Boston,First Edition,2017
2.	Modern power Electronics and AC Drives – B.K.Bose, Pearson Education India; 1st edition-2015

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3209	PE	3	--	--	3	25	75	3 Hrs.
SWITCHED MODE POWER SUPPLIES								
(For EEE)								
Course Objectives: student will learn								
1.	To study modern power semiconductor devices and its characteristics.							
2.	To Analyze and design Switched mode DC – DC converters for various applications.							
3.	To Comprehend the concepts of different Isolated DC – DC converters and their applications.							
4.	To Describe the importance of Resonant converters for industrial applications.							
5.	To study and analyze various types of inverters and its switching techniques.							
Course Outcomes:								
S.No	Student will be able to							Knowledge Level
1.	Assign suitable power semiconductor devices for particular converter topology.							K3
2.	Investigate the suitable filters for DC – DC converters.							K3
3.	Analyze various Isolated DC-DC converters.							K4
4.	Apply the resonance concept for DC-DC converters.							K3
5.	Examine various switching mode DC-AC converters.							K4
SYLLABUS								
UNIT-I (8 Hrs)		Power Semiconductor Devices: Modern power semiconductor devices - MOS turn Off Thyristor (MTO) —Emitter Turn off Thyristor (ETO) —Integrated Gate Commutated Thyristor (IGCTs) —MOS-Controlled Thyristors (MCTs) —Static Induction circuit —comparison of their features.						
UNIT-II (10 Hrs)		DC-to-DC Converters: Switching mode regulators: Buck, Boost, Buck Boost, Cuk regulators –Continuous Conduction mode – design of filter inductance & capacitance – boundary between continuous and discontinuous conduction – critical values of inductance/load resistance – Output voltage ripple.						
UNIT-III (12 Hrs)		Isolated DC-DC Converters: Fly back converter - continuous & discontinuous conduction mode, forward converter, push pull converter, half bridge converter, and full bridge converter - continuous conduction mode.						
UNIT-IV (12 Hrs)		Resonant Converters: Basic resonant circuit concepts – series resonant circuit – parallel resonant circuit – Series-loaded (SRL) and Parallel-loaded (PLR) resonant converters - ZCS resonant-switch converters - ZVS resonant-switch converter – comparison of ZCS & ZVS Resonant Converters.						
UNIT-V (8 Hrs)		Switched Mode DC to AC converters: 1-phase square wave full-bridge inverter – square wave switching scheme – sine PWM						

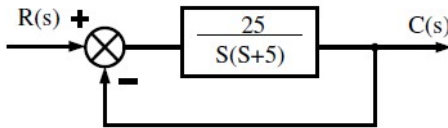
	switching scheme – PWM with bipolar & unipolar voltage switching – harmonic analysis of output voltage – Three phase sine PWM inverter.
Text Books:	
1.	Power Electronics: Circuits, devices and applications - M.H. Rashid- PHI., Third Edition.
2.	Power Electronics: Converters, Applications and Design - Ned Mohan, Tore M. Undel and, William P. Robbins - John Wiley & Sons., Third Edition
3.	Power Electronics- M.D. Singh and K. B. Khanchandani, Tata McGraw-Hill., Second Edition
Reference Books:	
1.	Power Electronics - P.S. Bimbhra, Khanna Publishers., Fifth Edition.
2.	Abraham Pressman, Switching Power supply Design, McGraw Hill
3.	Power Electronics by Daniel W Hart-Mc Graw Hill Publications, 1st Edition.
Web sources:	
1.	https://nptel.ac.in/courses/108/108/108108036/#

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3210	PE	--	--	--	3	25	75	3 Hrs.
MOOCS-III								
(For EEE)								
Course Objectives:								
MOOCs-III course should belong to the B.Tech. Programme concerned and that course should not be studied earlier. Students should select a course from SWAYAM/ NPTEL with minimum 12 weeks of duration. The percentage obtained for the candidate in MOOCs will be mapped to the grade table given in the Academic Regulations.								

Code	Category	L	T	P	C	I.M	E.M	Exam
#OE-I	OE	3	--	--	3	25	75	3 Hrs.
OPEN ELECTIVE-I								
(For EEE)								
Student has to study one Open Elective offered by CE or CSE or ECE or IT or ME or EM&H from the list enclosed.								

Offered by	Course Code	Course Name	Offered to
CIVIL ENGINEERING	B19CEOE01	Disaster Management	CSE,ECE,EEE, IT &ME
	B19CEOE02	Remote Sensing &GIS	
COMPUTER SCIENCE & ENGINEERING	B19CSOE01	Data Structures	CE,ECE,EEE, &ME
	B19CSOE02	Java Programming	
	B19CSOE02	Web Technologies	
ELECTRONICS & COMMUNICATIONENGINEERING	B19ECOEO1	Basic Electronics	CE,CSE,EEE, IT &ME
	B19ECOEO2	Signals & Systems	
INFORMATION TECHNOLOGY	B19ITOE01	Data Structures & Algorithms	CE,ECE,EEE, &ME
	B19ITOE02	Database Design & Development	
	B19ITOE03	Java Programming	
MECHNAICALENGINEERING	B19MEOE01	Operations Research	CE, CSE, ECE, EEE & IT
	B19MEOE02	Operations Management	
	B19MEOE03	Total Quality Management	
ENGINEERING MATHEMATICS & HUMANITIES	B19BSOE01	computational statistics with R	ALL BRANCHES
	B19BSOE02	Fuzzy sets and Fuzzy Logic	

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3211	PC	--	--	3	1.5	20	30	3 Hrs.
ELECTRICAL MACHINES - II LAB								
(For EEE)								
Course Objectives: Students will learn								
1.	Regulation of an alternator by using EMF, MMF and ZPF methods							
2.	A test on line excited induction generator and synchronous motor using different starting methods to draw V and inverted V curves.							
3.	No – Load and Blocked Rotor tests on a 3- Φ and 1- Φ induction motor.							
4.	A test on synchronous machine to find X_d and X_q and sequence reactance's.							
5.	Load test on 3- Φ induction motor and speed control on induction motor by pole changing method.							
Course Outcomes: Students will be able to								
S.No	Outcome:							Knowledge Level
1.	Conduct OC and SC tests on an alternator to Analyze the regulation of an alternator by using EMF, MMF and ZPF methods.							K4, K5
2.	Conduct a test on a Line excited induction generator and synchronous motor by using induction start method and synchronizing method to Analyze its performance characteristics							K4, K5
3.	Conduct No – Load and Blocked Rotor tests on a 3- Φ and 1- Φ induction motor and to Draw equivalent circuit and analyze the performance characteristics of a 3- Φ and 1- Φ induction motor.							K4, K5
4.	Conduct a test on a synchronous machine to determine X_d & X_q and sequence reactance's.							K4, K5
5.	Conduct load test on a 3- Φ induction motor to Draw and Analyzes its performance characteristics and Conduct speed control by pole changing method							K4, K5
LIST OF EXPERIMENTS								
1.	No – Load and Blocked Rotor test on a three-phase squirrel cage induction motor.							
2.	Regulation of an alternator by EMF and MMF methods.							
3.	Regulation of an alternator by ZPF method.							
4.	Characteristics of line excited induction generator.							
5.	Characteristics of induction start synchronous motor.							
6.	Load test on three phase slip ring induction motor.							
7.	V and inverted V curves of synchronous motor.							
8.	Measurement of X_d and X_q of a synchronous machine.							
9.	Equivalent circuit of a single-phase induction motor							
10.	Measurement of sequence reactances of a synchronous machine.							
11.	Speed Control of induction motor by pole changing method.							
Reference Books:								
1.	Nagrath& Kothari,” Electric Machines” TMH 5 th edition							
2.	PS Bhimbra, “Electrical Machinery”, Khanna Publishers. 7 th edition.							
3.	Fitzgerald & Kingsley, “Electric Machinery” McGraw Hill 6 edition.							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3212	PC	--	--	3	1.5	20	30	3 Hrs.
CONTROL SYSTEMS LAB								
(For EEE)								
Course Objectives: Student will								
1.	Understand the basic knowledge on practical control system.							
2.	Acquire knowledge on applications of machines & electronic devices with control systems.							
Course Outcomes: Student will be able								
1.	Formulate transfer function for given control system problems.							K4
2.	Find time response of given control system model.							K3
3.	Apply Root Locus and Bode plots for given control system model							K3
4.	Design Lead, Lag, Lead-Lag and PID controllers for given control system model							K5
LIST OF EXPERIMETNS								
1	Magnetic amplifier.							
2	Study of DC Servo motor.							
3	DC Position control system.							
4	Study of first order system.							
5	Study of second order system.							
6	Speed torque characteristics of AC Servomotor.							
7	PID Controller.							
8	Synchro Transmitter and Receiver pair.							
9	Study of digital control system.							
10	Study of frequency response of RC filter.							
Add-On Experiments								
11.	Speed- torque characteristics of Stepper Motor.							
12.	Transient response specifications of a given system for unit step input using MATLAB software.							
13	Design of a state feedback controller using MATLAB software for the given system to locate closed loop poles at -2+j4, -2-j4 and -10.					$\dot{X} = AX + Bu$ $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -5 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ the		
Reference Books:								
1.	Benjamin C. Kuo, „Automatic Control Systems“, PHI (5 th Edition).							
2.	Katsuhiko Ogata, „Modern Control Engineering“, PHI (4 th Edition).							
3.	Richard C. Dorf and Robert H. Bishop, „Modern Control Systems“, Addison-Wesley							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19EE3213	PC	--	--	3	1.5	20	30	3 Hrs.
POWER ELECTRONICS LAB								
(For EEE)								
Course Objectives: Students will learn								
1.	To realize the necessity of energy conversion and efficient implementation techniques.							
2.	To provide a practical exposure to operating principles, design, and synthesis of different power electronic converters.							
Course Outcomes: Student will be able to								
S.No	Outcome							Knowledge Level
1.	Conduct the experiment for the operation and characteristics of power semiconductor devices, Triggering circuits and powerconverters.							K5
2.	Conduct the experiment for the effect of Freewheeling diode on single-phase semi converter with RL-load & single-phase full converter with DC motor.							K5
3.	Conduct the experiment for the AC output voltage of AC voltage controller & Cycloconverter with changing frequency modes							K5
4.	Conduct the experiment for the performance of various Inverters.							K5
5.	Conduct the experiment for the performance of DC Choppers.							K5
LIST OF EXPERIMENTS								
1.	Study the Characteristics of SCR, IGBT and MOSFET							
2.	R and RC triggering circuits							
3.	Single Phase Semi Converter							
4.	Speed Control of DC Motor Using Single Phase Full Converter							
5.	Single Phase AC Voltage Controller							
6.	Single Phase Cycloconverter							
7.	DC Jones Chopper							
8.	Mc Murray Bedford Inverter							
9.	Single Phase Dual Converter							
10.	Three Phase Inverter (180 ⁰ and 120 ⁰ mode)							
11.	Three Phase Full Converter							
12.	Three Phase Semi Converter							
Reference Books:								
1.	Power Electronics: Circuits Devices and Applications – M.H. Rashid, Prentice Hall of India, 3 rd edition.							
2.	Power electronics - P.S. Bimbhra- Khanna Publishers, 4th Edition							

Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3201	MC	3	--	--	--	--	--	--
EMPLOYABILITY SKILLS II								
Common to All Branches								
Part-A: Verbal and Soft Skills-II								
Course Objectives:								
1.	To expose the students to bettering sentence expressions and also forming equivalents.							
2.	To instill reading and analyzing techniques for better comprehension of written discourses							
3.	To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.							
4.	To inculcate conversational skills, nuances required when interacting in different situations							
5.	To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.							
Course Outcomes: The students will be able to								
S.No	Outcome							Knowledge Level
1	Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.							K3
2	Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences							K3
3.	Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).							K3
4.	Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.							K3
5.	Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.							K3
SYLLABUS								
UNIT-I	Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.							
UNIT-II	Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author,,s intention of writing a passage), strategies for quick reading(importance given to skimming, scanning)							

	summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.
UNIT-III	Punctuation, discourse markers, general Essay writing, writing Issues and Arguments (with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a Statement of purpose Letters of Recommendation, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.
UNIT-IV	Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.
UNIT-V	Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.
Reading/ Listening material:	
1.	Guide to IELTS, Cambridge University Press.
2.	Barron,s GRE guide
3.	Newspapers like The Hindu, Times of India, Economic Times.
4.	Magazines like Frontline, Outlook and Business India.
5.	News channels NDTV, National News, CNN
Text Books:	
1.	Objective English and Verbal Reasoning by R S Agarwal
2.	Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP
3.	Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4.	Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5.	Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi.
Reference Books:	
1.	Oxford Guide to Effective Writing and Speaking by John Seely.
2.	Collins Cobuild English Grammar by Collins
3.	The Art of Public Speaking by Dale Carnegie
4.	The Leader in You by Dale Carnegie
5.	Emotional Intelligence by Daniel Golman
6.	Stay Hungry Stay Foolish by Rashmi Bansal.
Part-B: Quantitative Aptitude-II	
Course Objectives: The objective of introducing quantitative aptitude –II is:	

1.	To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages
2.	To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3.	To familiarize the students with finding out accurate date and time related problems
4.	To enable the students solve the puzzles using logical thinking.
5.	To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

S.No.	Course Outcome	Knowledge Level
1.	The students will be able to perform well in calculating different types of data interpretation problems.	K3
2.	The students will perform efficaciously on analytical and logical problems using various methods.	K3
3.	Students will find the angle measurements of clock problems with the knowledge of calendars and clock.	K3
4.	The students will skillfully solve the puzzle problems like arrangement of different positions.	K3
5.	The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.	K3

SYLLABUS

UNIT-I	Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.
UNIT-II	Puzzle test, blood Relations, permutations, Combinations and probability, Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.
UNIT-III	Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.
UNIT-IV	Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles
UNIT-V	Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals- Problems on quadrilaterals, Congruent triangles and properties, Similar triangles and its applications,

	Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.
Text Books:	
1.	Quantitative aptitude by RS Agarwal
2.	Verbal and non verbal reasoning by RS Agarwal. 3. Puzzles to puzzle you by shakunataladevi
3.	Puzzles to puzzle you by shakunataladevi
4.	More puzzles by shakunataladevi
5.	Puzzles by George summers
Reference Books:	
1.	Barron,,s by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2.	Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3.	Material from IMS, Career Launcher and Time,, institutes for competitive exams
4.	Books for CAT by arunsharma
5.	Elementary and Higher algebra by HS Hall and SR knight.

Subject Code	Category	L	T	P	C	I.M	E.M	Exam
B19MC3203	MC	--	--	3	0	--	--	--
ADVANCED CODING								
(Common to ECE & EEE)								
Course Objectives:								
1.	To understand the basics of Java Packages							
2	To learn about Collection Framework							
3	To investigate different methods of GUI Programming.							
4	To learn about Java collections and Libraries							
Course Outcomes: By the end of the course, the student should have the ability to:								
S.No	Outcome							Knowledge Level
1.	Able to solve problems using java collection framework and I/o classes.							K3
2.	Able to develop multithreaded applications with synchronization.							K3
3.	Able to develop applets for web applications							K3
4.	Able to design GUI based applications							K3
SYLLABUS								
UNIT-I (10 Hrs)	Standard Library templates and Java collections Introduction to Java as Object Oriented language, Essential Java Packages, Coding logics.							
UNIT-II (10 Hrs)	The Collections Framework (java.util)- Collections overview, Collection Interfaces, The Collection classes- Array List, Linked List, Hash Set, Tree Set, Priority Queue, Array Deque. Accessing a Collection via an Iterator, Using an Iterator, The For-Each alternative, Map Interfaces and Classes, Comparators, Collection algorithms, Arrays, The Legacy Classes and Interfaces- Dictionary, Hashtable ,Properties, Stack, Vector.							
UNIT-III (10 Hrs)	More Utility classes, String Tokenizer, Bit Set, Date, Calendar, Random, Formatter, Scanner.							
UNIT-IV (8 Hrs)	GUI Programming with Swing – Introduction, limitations of AWT, MVC architecture, components, containers. Understanding Layout Managers, Flow Layout, Border Layout, Grid Layout, Card Layout, Grid Bag Layout. Event Handling- The Delegation event model- Events, Event sources, Event Listeners, Event classes, Handling mouse and keyboard events.							
UNIT-V (12 Hrs)	Adapter classes, Inner classes, Anonymous Inner classes. A Simple Swing Application, Applets – Applets and HTML, Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet, Painting in Swing, A Paint example, Exploring Swing Controls- JLabel and Image Icon, JText Field, The Swing Buttons- JButton, JToggle Button, JCheck Box, JRadio Button, JTabbed Pane, JScroll Pane, JList, JCombo Box, Swing Menus, Dialogs.							
References:								
1.	Java The complete reference, 9th edition, Herbert Schildt, McGraw Hill Education (India) Pvt.							

	Ltd.
2.	Understanding Object-Oriented Programming with Java, updated edition, T. Budd, Pearson Education
3	An Introduction to programming and OO design using Java, J. Nino and F.A. Hosch, John Wiley & sons
4	Introduction to Java programming, Y. Daniel Liang, Pearson Education.
5	Object Oriented Programming through Java, P. Radha Krishna, University Press.
6	Programming in Java S. Malhotra, S. Chudhary, 2nd edition, Oxford Univ. Press.
7	Java Programming and Object-oriented Application Development, R. A. Johnson, Cengage Learning.
8	https://www.geeksforgeeks.org/
9	https://www.tutorialspoint.com/